

Momentum bookkeeping IDENTITY residual (not a conservation test)

Escaped trajectories

40

35

30

25

20

15

10

5

0

-14

-12

-10

-8

-6

-4

-2

0

$\log_{10}(|\Delta(P_N + P_{\text{rad}})|/P_{\text{scale}})$

Entries: N = 40

Fit: degenerate Gaussian ($\sigma=0$)

$\mu = -13.847$

$\sigma = 0$

IDENTITY by construction: E_{bound} is defined

as the residual that closes this sum;

this measures roundoff + endpoint interpolation.

Independent physical residual:

$|E_{\text{bound}}|/E_{\text{rad}}$ median/p95 = 65.276 / 90.295